

Epicyclic Gear System Analysis Tool

README / User Guide

MATLAB GUI + solver for reduced-order planetary gear load-sharing analysis

Files included in the project

- | | |
|---|----------------------------|
| • <code>launchEpicyclicGearAppV4.m</code> | Main launcher |
| • <code>EpicyclicGearAppV4.m</code> | Graphical user interface |
| • <code>EpicyclicGearSystem.m</code> | Numerical backend / solver |

How to run the tool

1. Keep all three files in the same folder.
2. Open that folder in MATLAB.
3. Run:

```
launchEpicyclicGearAppV4
```

4. The GUI will open automatically.

1. What the tool does

This tool models a planetary / epicyclic gear system using a reduced-order quasi-static load-sharing equilibrium model. Its main outputs are:

- planet load-sharing factors,
- the global load-sharing index K_γ ,
- floating sun displacement,
- tilt of the sun gear,
- phase-dependent responses,
- optional sensitivity and Monte Carlo studies.

2. General GUI workflow

Basic workflow

1. First choose or develop the desired **input parameters**.
2. Leave optional checkboxes **unchecked** if standard/default values are desired.
3. Check a box only when you want that model feature to become active and its values to affect the solution.
4. Press **RUN ANALYSIS**.
5. Read results in the summary panel, tables, and plots.

3. What each part of the GUI does

GUI area	Purpose
Run Analysis	Executes the solver using the current GUI inputs.
Open all plots in separate figures	Opens the same plots shown inside the GUI as separate MATLAB figure windows.
View Backend Code	Opens a read-only viewer containing the backend solver code (EpiccyclicGearSystem.m).
Basic Inputs	Defines the primary system settings: number of planets, input torque, sun radius, normal pressure angle, and bearing type.
Modify geometrical constraints?	Enables the geometry panel so the user can manually define tooth counts, module, helix angle, center distance, face width, profile shifts, and rigid offsets.
Non-standard pressure angle?	Allows a custom analysis pressure angle to be used in the line-of-action projection model.
Show advanced effects?	Enables the advanced-model options: error preset, thermal effects, periodic stiffness, eccentricity, support stiffness, Monte Carlo, sensitivity, etc.
Error level	Selects fine / medium / coarse default tolerances for manufacturing and assembly errors.
Zero error override	Forces all stochastic and preset errors to zero.
Error on planet 1 only	Applies a user-defined equivalent error to planet 1 only. Useful for controlled comparison cases.
Enable periodic eccentricity	Activates a rotating periodic eccentricity contribution.
Enable periodic stiffness	Activates a sinusoidally varying mesh stiffness model.
Enable thermal effects	Activates thermal stiffness scaling and, if enabled in the solver, thermal geometry mismatch.
Tilt support / Settlement support	Adds optional support stiffness on the generalized sun displacement coordinates.
Run sensitivity study	Performs parameter sweeps for eccentricity, mesh scale, and bearing scale.
Run Monte Carlo	Repeats the calculation over multiple seeded error realizations to study response variability.
Summary panel	Gives the main numerical results for the current run.
Final LSF / Error plots	Shows worst-phase load-sharing and error breakdowns.
System schematic	Displays a qualitative top-view schematic with sun, planets, eccentricity vector, and tilt direction.
Phase response	Shows K_γ as a function of phase angle over one cycle.
Sensitivity / Monte Carlo tabs	Shows statistical and parametric-response plots when enabled.

4. Literature-inspired run

The following settings reproduce the intended **literature-style comparison case** used in this project.

Recommended “Literature Run” settings

- Number of planets: **5**
- Input torque: **1000 N·m**
- Sun radius: **67.83 mm**
- Bearing type: **standard**
- Helix angle β : **13.12°**
- $z_1 = \mathbf{73}$, $z_2 = \mathbf{26}$, $z_3 = \mathbf{125}$
- Module $m_0 = \mathbf{1.81\text{ mm}}$
- Center distance $a_w = \mathbf{92.12\text{ mm}}$
- Face width $b = \mathbf{25\text{ mm}}$
- Individual planet error: **70 μm**
- No periodic stiffness
- No periodic eccentricity
- No thermal effects
- No random errors
- No sensitivity study
- No Monte Carlo
- Tilt support: **0**

Important note on pressure angle in the current GUI.

The current interface separates:

- the **normal pressure angle** (basic dropdown), and
- the **custom analysis / line-of-action pressure angle** (geometry panel checkbox + custom field).

For the literature-style case used in this project, the intended setup is:

- Basic dropdown (normal pressure angle): **20°**
- Check **Non-standard pressure angle?**
- Custom alpha: **23.04°**

How to set this run in the GUI

1. Set the basic inputs:

- Planets = 5
- Torque = 1000
- Sun radius = 67.83
- Normal pressure angle = 20
- Bearing type = standard

2. Check **Modify geometrical constraints?**

3. Set:

- $z_1 = 73$, $z_2 = 26$, $z_3 = 125$
- $m_0 = 1.81$
- $\beta = 13.12$

- $a_w = 92.12$
 - $b = 25$
 - Check **Non-standard pressure angle?**
 - Custom alpha = 23.04
4. Check **Show advanced effects?**
5. Set:
- Error on planet 1 only = checked
 - Individual planet error = 70
 - Periodic eccentricity = off
 - Periodic stiffness = off
 - Thermal effects = off
 - Zero error override = off
 - Tilt support = 0
 - Run sensitivity study = off
 - Run Monte Carlo = off
6. Press **RUN ANALYSIS**.

5. Main mathematical structure of the model

5.1 Generalized coordinates

The floating sun is represented by the reduced-order generalized displacement vector

$$\mathbf{q} = \begin{bmatrix} w \\ \phi_x \\ \phi_y \end{bmatrix}$$

where:

- w = settlement / approach,
- ϕ_x = tilt about the x -axis,
- ϕ_y = tilt about the y -axis.

5.2 Planet influence vector

For each planet at angular position θ_i , the influence vector is

$$\mathbf{v}_i = \begin{bmatrix} 1 \\ r \sin(\theta_i) \\ -r \cos(\theta_i) \end{bmatrix}$$

where r is the support radius used in the reduced-order tilt model.

5.3 Effective stiffness

The model combines sun-planet mesh stiffness, ring-planet mesh stiffness, and bearing stiffness through the equivalent stiffness relation

$$k_{\text{eff},i} = \left(\frac{1}{k_{SP,i} + k_{RP,i}} + \frac{1}{k_{B,i}} \right)^{-1}.$$

5.4 Quasi-static equilibrium

The connected equilibrium system is solved in the form

$$\mathbf{K}\mathbf{q} = \mathbf{g},$$

with

$$\mathbf{K} = \sum_i k_i \mathbf{v}_i \mathbf{v}_i^T + \mathbf{K}_{\text{support}},$$

and

$$\mathbf{g} = \mathbf{Q}_{\text{ext}} + \sum_i k_i s_i \mathbf{v}_i.$$

Here:

- k_i is the effective stiffness of planet i ,
- s_i is the equivalent non-negative gap for planet i ,
- $\mathbf{Q}_{\text{ext}}$ contains the nominal transmitted load and optional generalized moments.

5.5 Contact force

Once $\mathbf{q}$ is known, the spring compression for planet i is

$$\Delta_i = \mathbf{v}_i^T \mathbf{q} - s_i,$$

and the corresponding compression-only force is

$$F_i = k_i \Delta_i.$$

If $F_i \leq 0$, that planet is considered lifted off and removed from the active-contact solve.

5.6 Load sharing

The planet load share fraction is

$$\text{share}_i = \frac{F_i}{\sum_j F_j}.$$

The load-sharing factor of planet i is

$$LSF_i = \frac{F_i}{\frac{1}{N} \sum_j F_j}.$$

The overall system load-sharing index is

$$K_\gamma = \frac{\max(F_i)}{\frac{1}{N} \sum_j F_j}.$$

6. Error modelling

6.1 Real vs. equivalent errors

The code does not solve each real geometric deviation directly at the tooth-contact level. Instead, it converts them into an equivalent line-of-action (LOA) scalar error for each planet.

In other words:

- **real errors** = pin position errors, runout, assembly offsets, profile errors, thermal mismatch, etc.
- **equivalent errors** = a single scalar compatibility error per planet, projected onto the line of action.

6.2 Static equivalent error

The total static equivalent error for planet i is assembled as

$$e_{static,i} = e_{pin,i} + e_{runout,i} + e_{rigid,i} + e_{profile,i}.$$

6.3 Phase-dependent total error

When periodic effects are active, the total error becomes

$$e_{total,i}(\gamma) = e_{static,i} + e_{ecc,i}(\gamma) + e_{th,i},$$

where:

- $e_{ecc,i}(\gamma)$ is the periodic eccentricity contribution,
- $e_{th,i}$ is the thermal geometry mismatch contribution,
- γ is the phase variable over one cycle.

6.4 Gap definition

The equivalent gap vector is then defined as

$$s_i = \max_j(e_{total,j}) - e_{total,i}.$$

This ensures the most protruding planet has zero gap, while the others must close a compatibility gap before carrying load.

7. Thermal model

The thermal model is a reduced-order engineering approximation, not a full thermo-elastic tooth-contact analysis.

7.1 Thermal stiffness scaling

The mesh stiffnesses are scaled using

$$\begin{aligned}\Delta T &= T - T_{ref}, \\ E_{ratio} &= 1 - c_E \Delta T, \\ b_{ratio} &= 1 + \alpha \Delta T, \\ \text{scale} &= E_{ratio} b_{ratio}.\end{aligned}$$

This represents:

- a decrease in retained stiffness with temperature,
- a small geometric correction through thermal expansion of face width.

7.2 Thermal geometry mismatch

The local thermal field around the planets is modelled as

$$\Delta T_i = \Delta T_{mean} + \Delta T_{grad} \cos(\psi_i - \psi_{hot}),$$

so that the planets can experience slightly different local temperatures.

A reduced thermal geometry error is then built from:

- carrier/pin-circle thermal expansion,
- planet effective pitch-radius growth.

The final thermal error is combined as

$$e_{th,i} = w_{pin} e_{pin,th,i} + w_{planet} e_{planet,th,i},$$

and its common-mode part is removed:

$$e_{th,i} \leftarrow e_{th,i} - \overline{e_{th}}.$$

This means only the relative thermal mismatch between planets influences load sharing.

8. Important modelling assumptions

- The sun gear is treated as a floating member.
- The model is not fully dynamic, therefore it does not function relative to time, but relative to the rotation of the sun gear.
- Contact is compression-only, so lift-off is possible.
- Mesh and bearing stiffnesses are represented through equivalent reduced-order stiffnesses.
- Real geometry deviations are converted into equivalent line-of-action errors.

End of README